

Host workstation

Xilinx ISE bitstream build
+ JTAG programming

Registered replay generator
common request corpus

Fixed-point expected-output generator

UART capture + parser
retain raw QF6R record before parsing

Nine-field exact comparator
captured integers = ex-
pected integers; no tolerance

CSV / JSON result generation

Nexys 3 FPGA: independently built H0–H4, one at a time

Volatile H0, H1, H2,
H3, or H4 bitstream
100 MHz kernel clock

Synchronous request interface
accepted edge: start && ready

Candidate evaluator
fixed-point selection

Registered result + done edge
16-bit cycle counter

QF6R result formatter

115,200-baud UART

JTAG programming path

request / data input path

UART output path



Actual Nexys 3 Rev.
B board used for
replay (uncropped
photograph).

Excluded from reported kernel latency: JTAG programming, UART serialization,
host processing, parsing, candidate generation, KMS/SDN access, and route installation.

Physical evidence scale: 84 records per configuration; five independent config-
urations; 420 physical records; nine fields per record; 3,780 exact comparisons.